

CONVERSATIONAL CLUSTERING

Intelligent data clustering through natural language interaction with
Large Language Models (LLM)

The Executive Blind Spot

Executives (CEOs, Directors) need to extract insights from massive customer feedback (e.g., 20,000+ reviews).

-  **Volume:** 20,000+ reviews are unreadable manually.
-  **Skill Gap:** No ML expertise to tune algorithms.
-  **Need:** Rapid identification of complaints vs. praise.



Conversational Intelligence



Natural Interface

A chat platform where users clarify clustering goals with an LLM agent.



Automated ML

The LLM autonomously selects optimal algorithms and parameters.



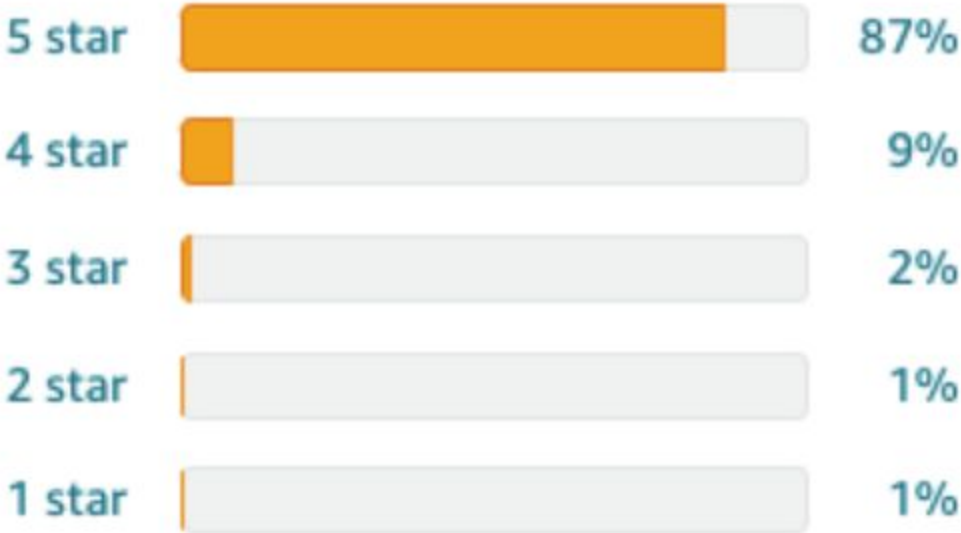
Visual Action

Returns clear cluster structures, statistics, and concrete examples.

Amazon Reviews Dataset

A high-impact foundation for e-commerce sentiment analysis at scale.

-  **20,000+** Records
-  Rich metadata & text titles
-  Global customer insights



▼ [How are ratings calculated?](#)

By feature

Durability	★★★★★	4.9
Value for money	★★★★★	4.8
Giftable	★★★★★	4.8
Craftsmanship	★★★★★	4.8

SOTA Research Workflow

Text Clustering as Classification with LLMs

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Abstract

Text clustering serves as a fundamental technique for organizing and interpreting unstructured textual data, particularly in contexts where manual annotation is prohibitively costly. With the rapid advancement of Large Language Models (LLMs) and their demonstrated effectiveness across a broad spectrum of NLP tasks, an emerging body of research has begun to explore their potential in the domain of text clustering. However, existing LLM-based approaches still rely on fine-tuned embedding models and sophisticated similarity metrics, rendering them computationally intensive and necessitating domain-specific adaptation. To address these limitations, we propose a novel framework that reframes text clustering as a classification task by harnessing the in-context learning capabilities of LLMs. Our framework eliminates the need for fine-tuning embedding models or intricate clustering algorithms. It comprises two key steps: first, the LLM is prompted to generate a set of candidate labels based on the dataset and then merges semantically similar labels; second, it assigns the most appropriate label to each text sample. By leveraging the advanced natural language understanding and generalization capabilities of LLMs, the proposed approach enables effective clustering with minimal human intervention. Experimental results on diverse datasets demonstrate that our framework achieves comparable or superior performance to state-of-the-art embedding-based clustering techniques, while significantly reducing computational complexity and resource requirements. These findings underscore the transformative potential of LLMs in simplifying and enhancing text clustering tasks. We make our code available to the public for utilization¹.

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1 Introduction

Text clustering is a fundamental task in natural language processing (NLP), which aims to group similar texts based on their content without prior labeling. It is widely applied in scenarios where manual annotation is costly or impractical, such as improving community detection in social media [44, 58], identifying emerging topics [9, 42], analyzing large-scale textual datasets [2, 37], structuring unorganized information [13], and enhancing document retrieval [1, 3, 12]. Despite its importance, traditional text clustering faces notable challenges in both methodology and implementation.

Traditional text clustering methods typically involve transforming textual data into numerical representations and applying clustering algorithms to group similar texts based on these representations. A common approach is to use pre-trained embedding models [16, 38, 52, 57] to convert text into dense vector embeddings that capture semantic relationships. These embeddings are then clustered using algorithms like K-means [35], DBSCAN [18], or hierarchical clustering [27]. However, fine-tuning embeddings for domain-specific tasks is computationally expensive and requires labeled data [51]. Besides, clustering algorithms are sensitive to hyperparameters, such as the number of clusters and distance metrics, which often need manual tuning based on expert knowledge. The resulting clusters also lack interpretability, as clustering models typically do not produce meaningful labels for the groups. These challenges make traditional clustering approaches less flexible and efficient

Improving Hierarchical Text Clustering with LLM-guided Multi-view Cluster Representation for Interaction Drivers in Contact Centers

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Abstract

In this work, we present an approach that introduces different perspectives or views to improve the quality of hierarchical clustering of interaction drivers in a contact center. Specifically, we present a multi-stage approach that introduces LLM-guided multi-view cluster representation that significantly improves the quality of generated clusters. Our approach improves average Silhouette Score by upto 70% and Human Preference Scores by 36.7% for top-level clusters compared to standard agglomerative clustering for the given business use-case. We also present how the proposed approach can be adapted to cater to a standard non-hierarchical clustering use-cases where it achieves state-of-the-art performance on public datasets based on NMI and ACC scores, with minimal number of LLM queries compared to the current state-of-the-art approaches. Moreover, we apply our technique to generate two new labeled

of this data for downstream business use-cases is tedious and time-consuming.

Contact center interaction drivers are often thought of as having a two-level hierarchical structure consisting of a few main categories with several sub-categories under each main category. These are often referred to as level-1 (L1) and level-2 (L2) categories respectively. This makes it conducive to applying hierarchical clustering techniques to organize them into L1 & L2 clusters. However, while contact center business use cases call for the best quality especially at the top-level for L1 clusters, current state-of-the-art clustering techniques fall short of this. Specifically, L1 categories surfaced by existing methodologies often fail to bring out the right abstraction and multifaceted similarities within the L2 categories. Meanwhile, we notice that capturing this abstraction comes naturally to humans, and in recent times the best of

SOTA Research Workflow

LLM-Guided Lifecycle-Aware Clustering of Multi-Turn Customer Support Conversations

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Abstract

Clustering customer chat data is vital for cloud providers handling multi-service queries. Traditional methods struggle with overlapping concerns and create broad, static clusters that degrade over time. Re-clustering disrupts continuity, making issue tracking difficult. We propose an adaptive system that segments multi-turn chats into service-specific concerns and incrementally refines clusters as new issues arise. Cluster quality is tracked via Davies–Bouldin Index (DBI) and Silhouette Scores, with LLM-based splitting applied only to degraded clusters. Our method improves Silhouette Scores by over 100% and reduces DBI by 65.6% compared to baselines, enabling scalable, real-time analytics without full re-clustering.

1 Introduction

Cloud providers handle high volumes of customer

the structure and categorization of customer support conversations in cloud service environments. Within each group, HDBSCAN + UMAP (McInnes et al., 2017, 2018a,b) creates interpretable topic clusters, labeled using LLMs (Ma et al., 2024; Pattnayak et al.). New concerns are incrementally matched; unmatched ones form new clusters when volume permits.

We track DBI and Silhouette Scores (Davies and Bouldin, 1979; Rousseeuw, 1987a). to monitor cluster quality. Degraded clusters are flagged using Z-score and cohesion tests, then refined using LLM-based splitting avoiding full re-clustering and preserving cluster identity for stable, actionable insights.

Our Contributions:

- LLM-based segmentation of multi-turn

NILC: Discovering New Intents with LLM-assisted Clustering

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Abstract

New intent discovery (NID) seeks to recognize both new and known intents from unlabeled user utterances, which finds prevalent use in practical dialogue systems. Existing works towards NID mainly adopt a cascaded architecture, wherein the first stage focuses on encoding the utterances into informative text embeddings beforehand, while the latter is to group similar embeddings into clusters (i.e., intents), typically by *K*-Means. However, such a cascaded pipeline fails to leverage the feedback from both steps for mutual refinement, and, meanwhile, the embedding-only clustering overlooks nuanced textual semantics, leading to suboptimal performance.

To bridge this gap, this paper proposes NILC, a novel clustering framework specially catered for effective NID. Particularly, NILC follows an iterative workflow, in which clustering assignments are judiciously updated by carefully refining cluster centroids and text embeddings of uncertain utterances with the aid of *large language models* (LLMs). Specifically, NILC first taps into LLMs to create additional *semantic centroids* for clusters, thereby enriching the contextual semantics of the Euclidean centroids of embeddings. Moreover, LLMs are then harnessed to augment hard samples (ambiguous or terse utterances) identified from clusters via rewriting for subsequent cluster correction. Further, we inject supervision signals through non-trivial techniques *seeding* and *soft must links* for more accurate NID in the semi-supervised setting. Extensive experiments comparing NILC against multiple recent baselines under both unsupervised and semi-supervised settings showcase that NILC can achieve significant performance improvements over six benchmark datasets of diverse domains consistently.

1 Introduction

New Intent Discovery (NID) [23] aims to identify and group user utterances from conversational systems into both known and previously unseen intents (goals) in an open-world context, which outputs new intent labels or heuristics for faster annotations thereafter and automates the expensive manual annotation of tremendous data. This task serves as building blocks to underpin a variety of applications, such as enhancing task-oriented dialogue systems (e.g., chatbots) [7] in e-commerce or other platforms [9, 10, 27], refining search results through better query understanding [38], and elevating personalized services [15, 16] with better user modeling. The ability to dynamically discover and adapt to new intents is essential to ensure the effectiveness of these systems in an open-world environment where user needs are constantly evolving.

Existing solutions towards NID tasks can be broadly divided into two categories: *embedding-based methods* and *LLM-driven approaches*. Embedding-based methods typically follow a cascaded pipeline. They first encode all the utterances into a semantic space (i.e., text embeddings), followed by applying standard clustering algorithms like *K*-Means to group utterances into intent clusters [1, 5, 44]. Most of them focus on pretraining or fine-tuning a text encoder, usually small language models, on available utterance corpora with limited or even without annotations, and thus, often struggle to capture the nuanced semantics hidden in utterances, particularly when dealing with technical or ambiguous utterances. Moreover, while this modular methodology enjoys merits in simplicity and flexibility, it fails to leverage the results from two independent steps to optimize each other.

With the advent of *large language models* (LLMs), a promising

SOTA Research Workflow

Large Language Models Enable Few-Shot Clustering

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Abstract

Unlike traditional unsupervised clustering, semi-supervised clustering allows users to provide meaningful structure to the data, which helps the clustering algorithm to match the user's intent. Existing approaches to semi-supervised clustering require a significant amount of feedback from an expert to improve the clusters. In this paper, we ask whether a large language model (LLM) can amplify an expert's guidance to enable query-efficient, few-shot semi-supervised text clustering. We show that LLMs are surprisingly effective at improving clustering. We explore three stages where LLMs can be incorporated into clustering: before clustering (improving input features), during clustering (by providing constraints to the clusterer), and after clustering (using LLMs post-correction). We find that incorporating LLMs in the first two stages rou-

2004). These interfaces have all been shown to give experts control of the final clusters. However, for large, real-world datasets with a large number of possible clusters, the feedback cost required by interactive clustering algorithms can be immense.

Building on a body of recent work that uses large language models (LLMs) as noisy simulations of human decision-making (Fu et al., 2023; Horton, 2023; Park et al., 2023), we propose a different approach for semi-supervised text clustering, illustrated in Figure 1. In particular, we answer the following research question: *Can an expert provide a few demonstrations of their desired interaction (e.g., pairwise constraints) to a large language model, then let the LLM direct the clustering algorithm?*

We explore three places in the text clustering

SOTA Research Workflow

IntentGPT: Few-shot Intent Discovery with Large Language Models

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Abstract

In today's digitally driven world, dialogue systems play a pivotal role in enhancing user interactions, from customer service to virtual assistants. In these dialogues, it is important to identify user's goals automatically to resolve their needs promptly. This has necessitated the integration of models that perform Intent Detection. However, users' intents are diverse and dynamic, making it challenging to maintain a fixed set of predefined intents. As a result, a more practical approach is to develop a model capable of identifying new intents as they emerge. We address the challenge of Intent Discovery, an area that has drawn significant attention in recent research efforts. Existing methods need to train on a substantial amount of data for correctly identifying new intents, demanding significant human effort. To overcome this, we introduce IntentGPT, a novel training-free method that effectively prompts Large Language Models (LLMs) to discover new intents.

involve seeking information, making requests, or expressing opinions, and steering the conversation appropriately. This task is dynamic since user intents evolve and new digital tools continually emerge, expanding the range of intents. Intent prediction needs to adapt to this open-world scenario and stay responsive to changing user needs.

Intent Detection has received considerable research in recent years (Lin et al., 2023; Costello et al., 2018; FitzGerald et al., 2022; Jayarao and Srivastava, 2018; Zhang et al., 2020). These works approach the problem from a closed-world perspective, with predefined classes, while often allowing for predictions on out-of-scope examples into an additional "other" class. Our focus lies in Intent Discovery (Shi et al., 2018; Zhang et al., 2022), which assumes an open-world setting where novel intents must be discovered. This approach aligns better with real-world applications.

CLUSTERLLM: Large Language Models as a Guide for Text Clustering

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Abstract

We introduce CLUSTERLLM, a novel text clustering framework that leverages feedback from an instruction-tuned large language model, such as ChatGPT. Compared with traditional unsupervised methods that builds upon "small" embedders, CLUSTERLLM exhibits two intriguing advantages: (1) it enjoys the emergent capability of LLM even if its embeddings are inaccessible; and (2) it understands the user's preference on clustering through textual instruction and/or a few annotated data. First, we prompt ChatGPT for insights on clustering perspective by constructing hard triplet questions *<does A better correspond to B than C>*, where A, B and C are similar data points that belong to different clusters according to small embedder. We empirically show that this strategy is both effective for fine-tuning small embedder and cost-efficient to query ChatGPT. Second, we prompt ChatGPT for help on

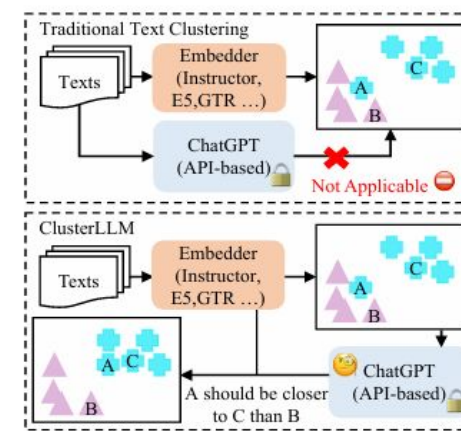


Figure 1: LLMs like ChatGPT are not applicable for text clustering directly because of the inaccessible embeddings. CLUSTERLLM resolves the dilemma by leveraging LLM as a guide on text clustering.

IntentGPT

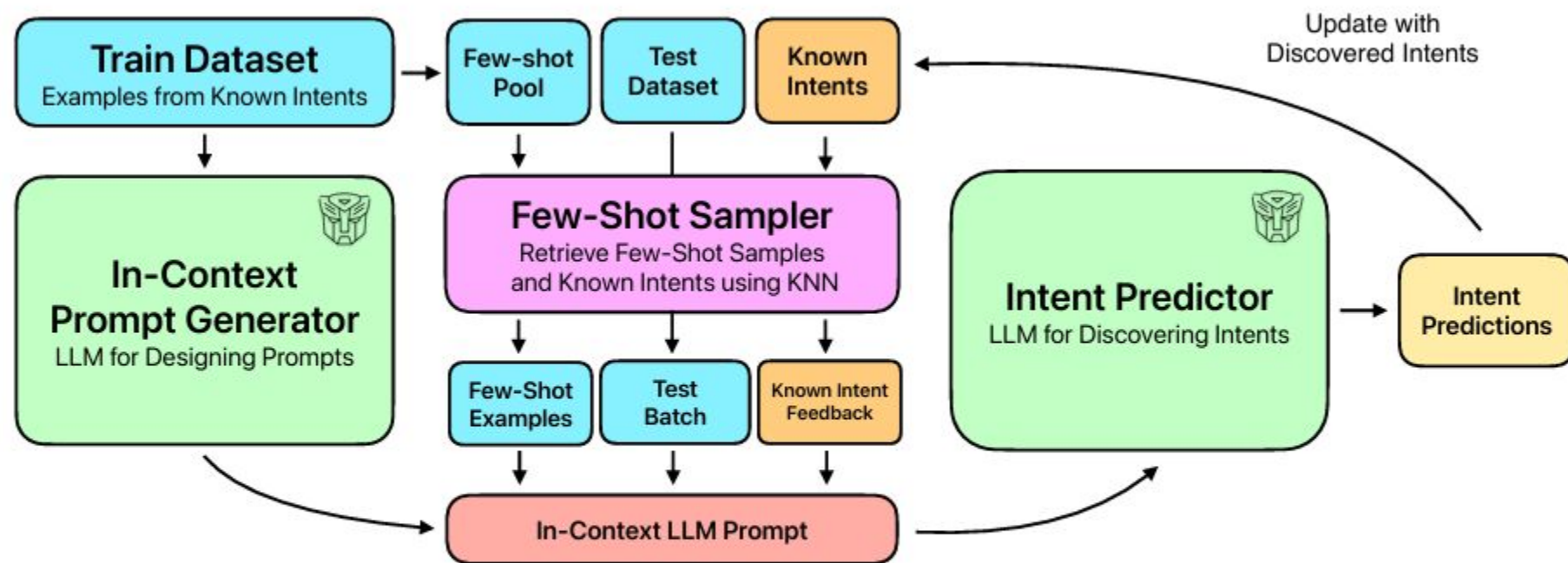


Figure 1: **IntentGPT architecture.** The system consists of two frozen LLMs. The In-Context Prompt Generator receives a small subset of training data and is devoted to generating a task prompt using domain context. The generated prompt is filled with, few-shot samples, a list of known intents, and the test sample (user query). Finally, the Intent Predictor assigns (or discovers) the intent and updates the Known Intents database accordingly.

CLUSTERLLM

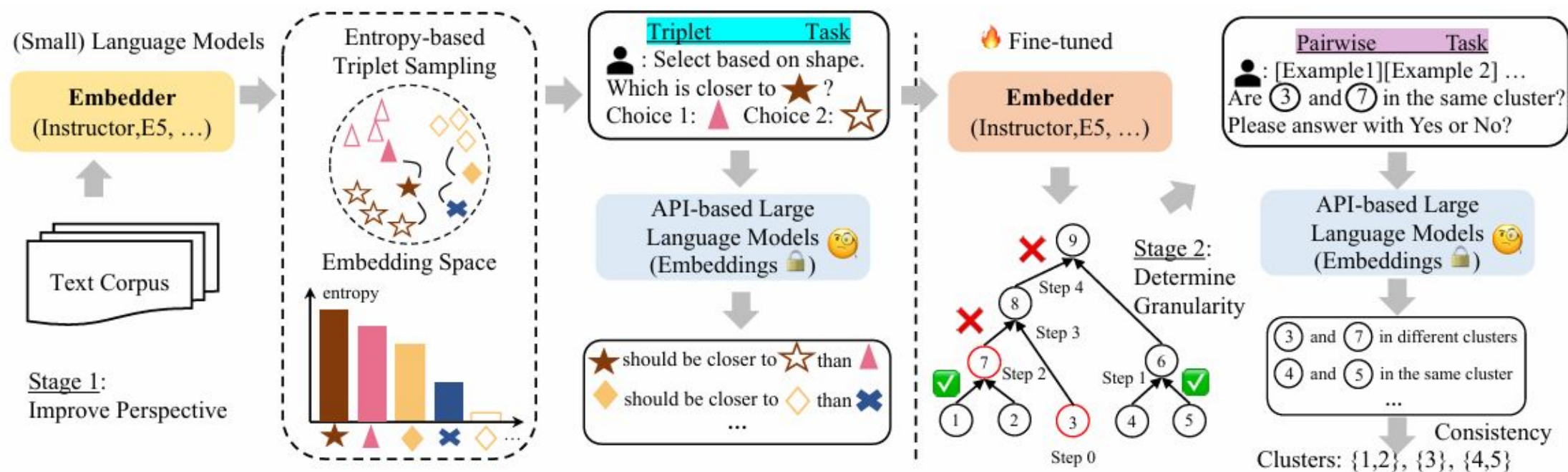


Figure 2: An overview of CLUSTERLLM. It utilizes LLM to guide an embedder for text clustering with a low cost.

Comparative Analysis: IntentGPT vs. ClusterLLM

IntentGPT

Pros

- **Training Free:** Operates without the need for extensive model fine-tuning.
- **Architectural Simplicity:** Fully automated execution by LLM reduces engineering overhead.

Cons

- **High Latency:** Processing can last several hours per dataset depending on model scale.
- **Prohibitive Cost:** Significant financial investment required for large-scale API usage.

ClusterLLM

Pros

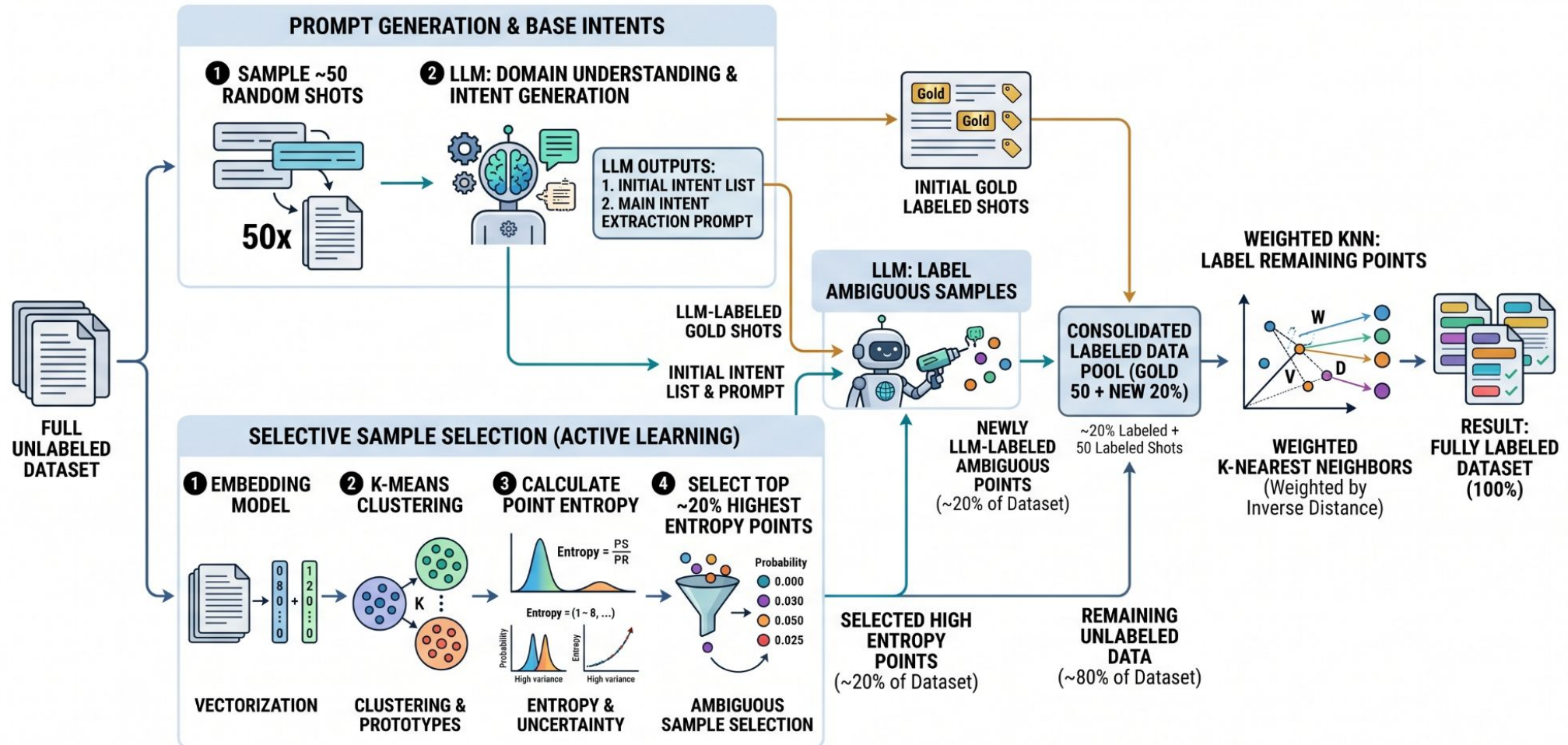
- **High Efficiency:** Rapid execution, processing datasets in just several minutes.
- **Cost-Effective:** Significantly cheaper to run per dataset (avg. ~\$0.6).

Cons

- **Hardware Dependency:** Requires dedicated GPU resources for embedding fine-tuning.
- **Complexity:** Architecture involves intricate multi-stage steps and heavy calculations.

Priority Research Insights

HYBRID INTENT DISCOVERY WORKFLOW: ACTIVE LEARNING & PROMPT GENERATION



Evaluation Strategies

✔Ground-Truth Required

NMI / ARI	Mutual Info & Rand Index
ACC	Clustering Accuracy
F1-Score	Precision/Recall Balance

❓Unsupervised (No Labels)

Silhouette	Separation & Density
DBI	Davies-Bouldin Index
Goodness	LLM-based Quality Score

Innovating Unsupervised Testing

LLM
as a Judge

The Goodness Score

In real-world business data, **ground-truth labels do not exist.**

-  LLM evaluates semantic consistency.
-  Automates the "Manual Review" process.
-  Provides actionable quality ratings.

Innovating Unsupervised Testing

Dial-In LLM: Human-Aligned LLM-in-the-loop Intent Clustering for Customer Service Dialogues

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Abstract

Discovering customer intentions is crucial for automated service agents, yet existing intent clustering methods often fall short due to their reliance on embedding distance metrics and neglect of underlying semantic structures. To address these limitations, we propose an **LLM-in-the-loop (LLM-ITL)** intent clustering framework, integrating the language understanding capabilities of LLMs into conventional clustering algorithms. Specifically, this paper (1) examines the effectiveness of fine-tuned LLMs in semantic coherence evaluation and intent cluster naming, achieving over 95% accuracy aligned with human judgments; (2) designs an LLM-ITL framework that facilitates the iterative discovery of coherent intent clusters and the optimal number of clusters; and (3) introduces context-aware techniques tailored for customer service dialogue. Since existing English benchmarks lack sufficient semantic diversity and intent coverage, we further present a comprehensive Chinese dialogue in-

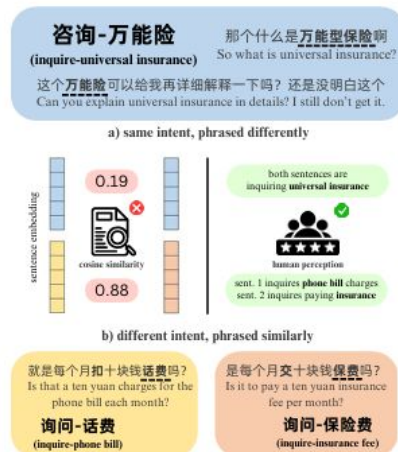


Figure 1: Comparison of embedding-distance metric and human perception: cosine similarity failed to identify the same intentions under diverse expression (top), and to distinguish distinct intentions under similar expressions (bottom).

3.1 LLM Utilities

Coherence Evaluation. Inspired by human behaviors in perceiving texts and making comparisons based on semantic meaning rather than surface-level word similarity (Peter W. Foltz and Landauer, 1998), this paper proposes **semantic coherence** as a more effective metric and optimization objective, measuring the semantic consistency within a cluster. The coherence evaluation is formulated as a binary classification problem, enabling intuitive interpretation and simplifying the training of the LLM evaluator (see Section 5.6 for alternative formulations). **Good** clusters consist of sentences focused on a specific topic, whereas **Bad** clusters contain inconsistent or ambiguous intentions (see Table 11 for examples). This task is particularly challenging for traditional machine learning classifiers due to their lack of semantic understanding (Mimno et al., 2011), thus necessitating fine-tuned

Questions?

Thank you for your attention!

Conversational Clustering Project